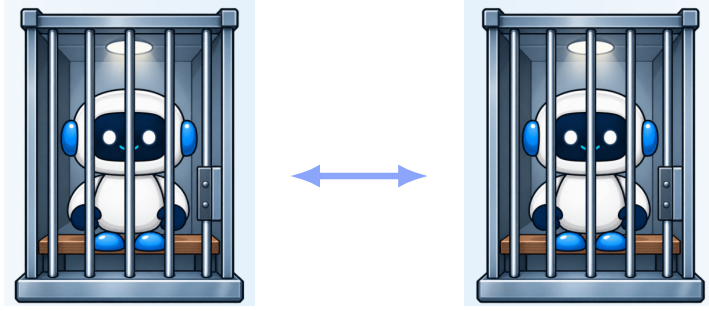
Payoff Scaling Shapes Cooperation in LLM Agents Across Languages

The Anh Han
Professor of Computer Science, Teesside University

Same game → **Different units ($\times\lambda$)**



Does an LLM agent play the same game when every payoff is multiplied by $\lambda > 0$?

Joint work with T.-K. Huynh, D.-S. Duy-Minh and 13 co-authors (HCMUS, HCMUT, VNU-HCM, LIST)

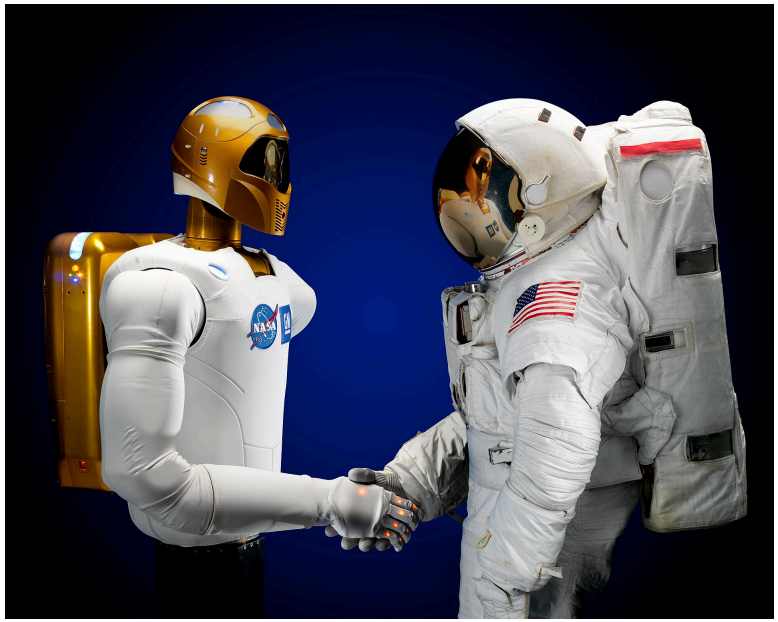
t.han@tees.ac.uk

Outline

1. **Introduction**
   - **LLM agents in cooperation dilemmas**
   - **Payoff scaling: same game, different units**
2. **Experimental design**
3. **Results**
   - **Cooperation rates**
   - **Strategies**
4. **Conclusions**

LLMs increasingly act as agents for us

- **LLMs negotiate, coordinate and decide on behalf of users**
- **Their behaviour emerges from strategic interaction**
- **They face cooperation dilemmas: safety, coordination, AI governance**



How do we measure LLM cooperation?

- **Let LLMs play classical games, e.g. the prisoner's dilemma**
- **Report one cooperation rate, under one payoff matrix**
- **Report what agents did, not the rule behind it**

Is a cooperation rate a property of the model, or of how the game is written down?

The (iterated) prisoner's dilemma

Cooperate
pays a cost so both do better

Defect
exploits, or avoids being exploited

	C	D
C	R, R	S, T
D	T, S	P, P

$$T > R > P > S$$

A strategy maps the history to the next move

- **AllC: always cooperate**
- **AllD: always defect**
- **TFT: copy the partner**
- **WSLS: win-stay, lose-shift**



Same game, different units

- **Multiply every payoff by $\lambda > 0$**
- **Preferences, best replies, Nash equilibria and replicator orbits are all unchanged**
- **Cents, dollars or thousands of pounds: the same game**



John von Neumann
von Neumann & Morgenstern, *Theory of Games*, 1944

Predicted effect of λ on a rational agent: exactly zero

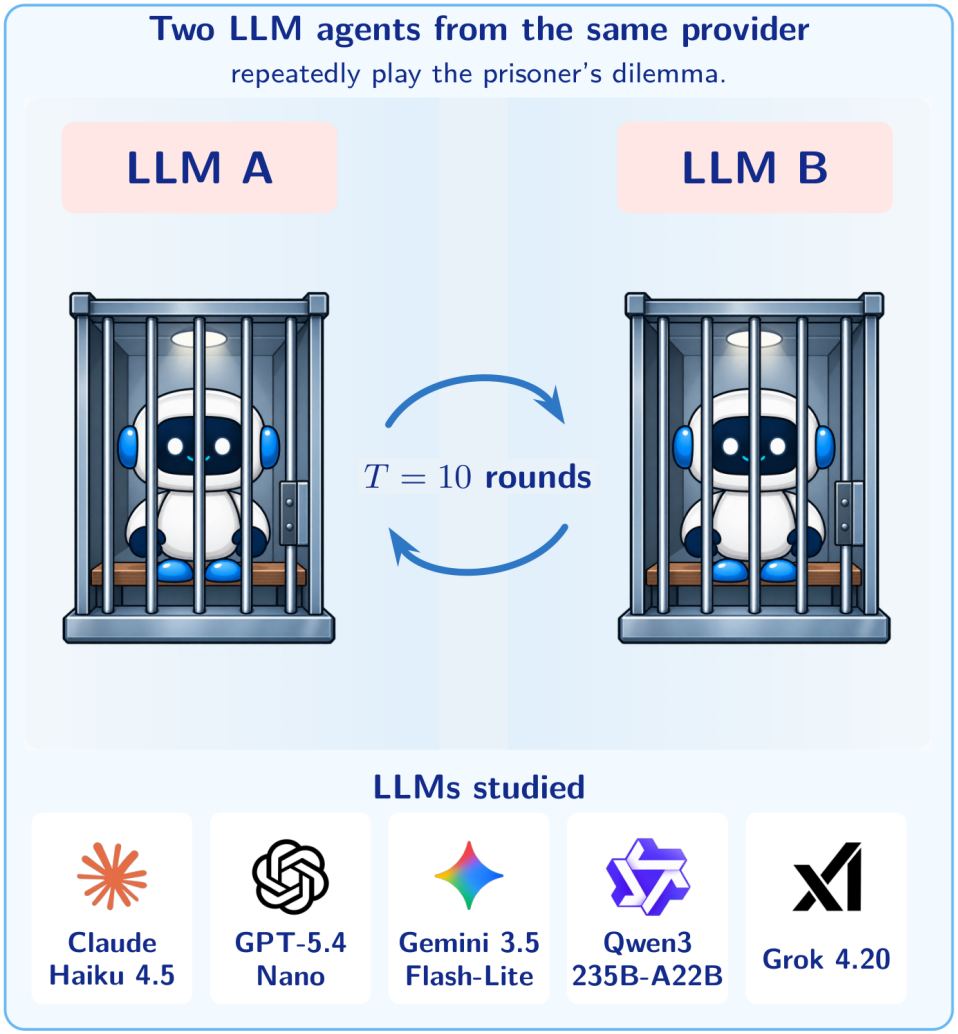
Research questions

1. **Is LLM behaviour invariant to rescaling the payoffs?**

2. **Does it depend on the language and the persona?**

3. **Does rescaling change only how often a model cooperates, or also its strategy?**

Experimental design



- Two agents of the same model play each other
- 10 rounds of the prisoner's dilemma
- Every payoff multiplied by λ

The payoff-scaled prisoner's dilemma

- **FAIRGAME[1] scenario: a criminal interrogation**



- **Penalties to minimise**

$\lambda = 0.1$

	A	B
A	0.6, 0.6	0, 1.0
B	1.0, 0	0.2, 0.2

$\lambda = 1$

	A	B
A	6, 6	0, 10
B	10, 0	2, 2

$\lambda = 10$

	A	B
A	60, 60	0, 100
B	100, 0	20, 20

- **Option A = defect, Option B = cooperate**

- **Ten scales, five orders of magnitude:**

$$\lambda \in \{0.01,\ 0.1,\ 0.25,\ 0.5,\ 1,\ 2,\ 5,\ 10,\ 100,\ 1000\}$$

[1] Buscemi, Proverbio, Di Stefano, Han, Castignani, Liò. **FAIRGAME: a framework for AI agents bias recognition using game theory**. ECAI, 2025.

Models, languages, personas

- **5 models, each playing itself**
  Claude Haiku 4.5, GPT-5.4 Nano, Gemini 3.5 Flash-Lite, Qwen3 235B-A22B, Grok 4.20

- **5 languages**
  English, French, Vietnamese, Chinese, Arabic

- **2 personas**
  “You are cooperative.”
  “You are selfish.”

- **10 rounds, simultaneous moves**

Hello

Bonjour

Xin chào

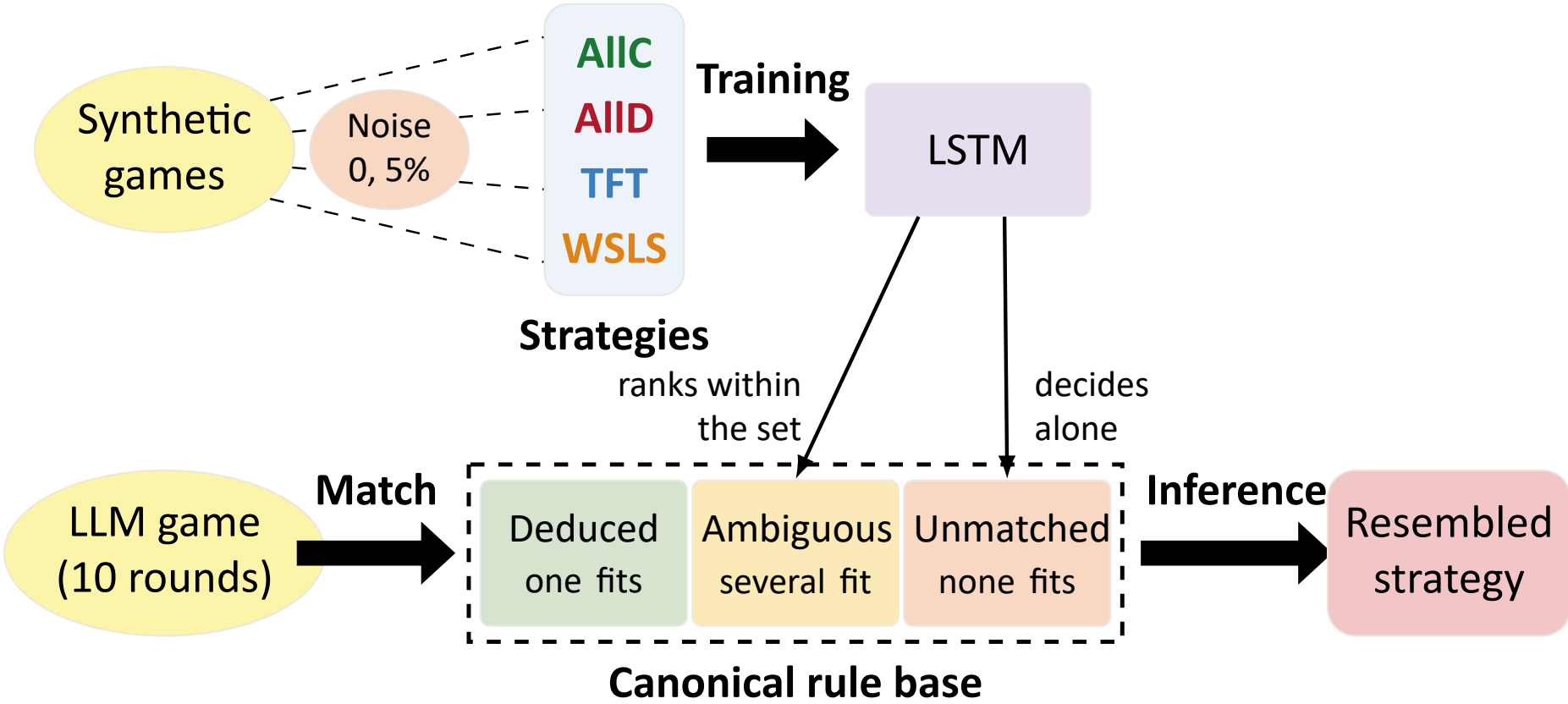
你好

مرحبا



10,000 games · 200,000 decisions

Strategy read-out[2]



Resembles, not plays

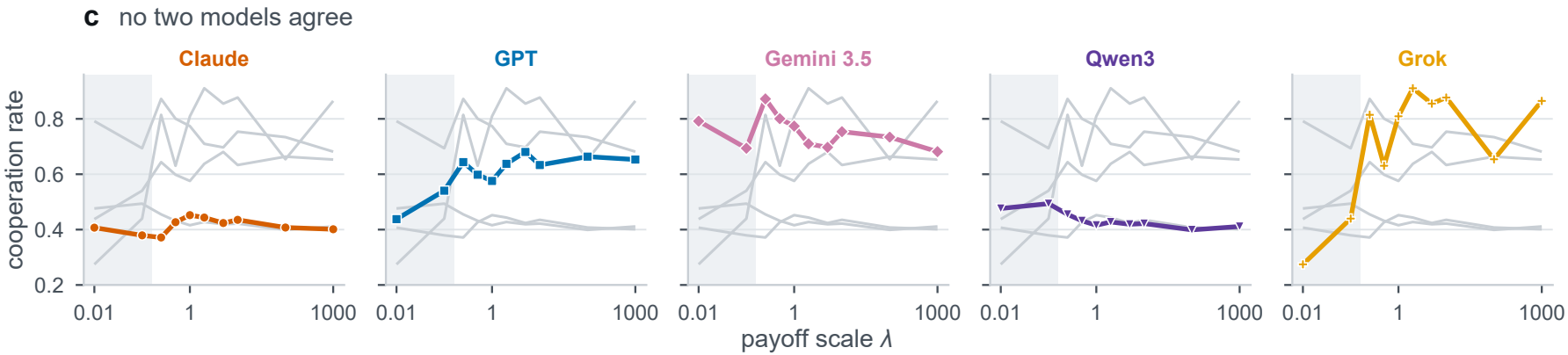
[2] Di Stefano, Jayne, Angione, Han. **Recognition of behavioural intention in repeated games using machine learning**. ALIFE, 2023.

Every model violates scale invariance



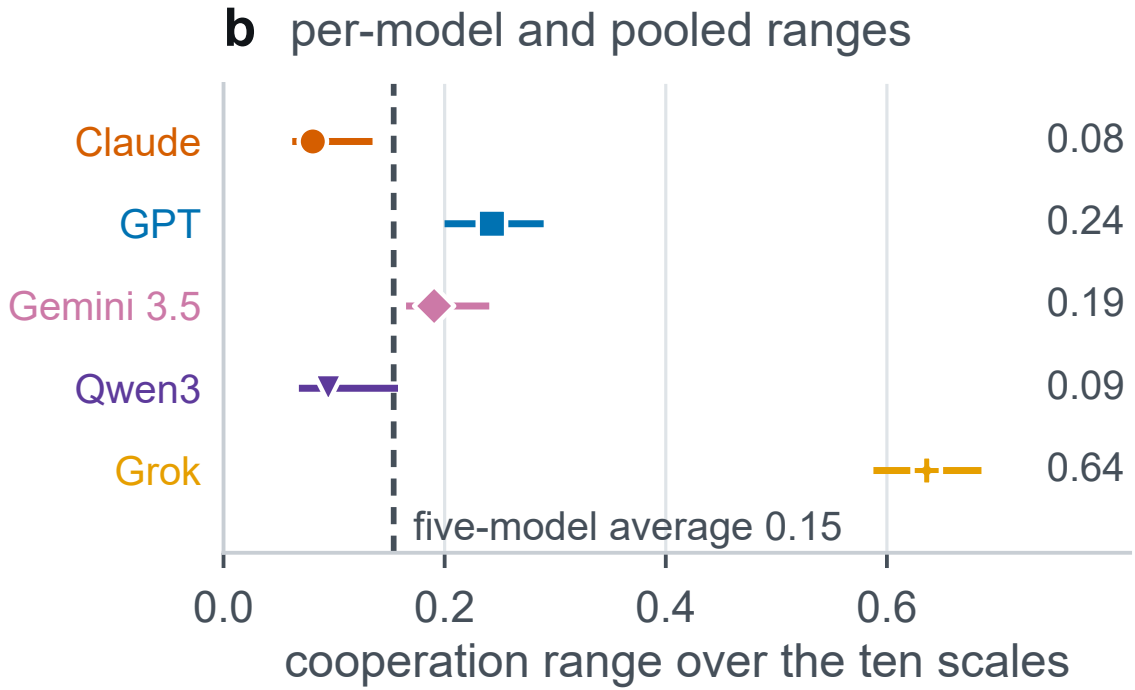
- **Theory predicts flat rows: every row moves**
- **Grok: 0.27 at $\lambda = 0.01$, 0.91 at $\lambda = 2$**

Five models, five machine characters



- **Up, down, dip, jump: each model has its own response**
- **No single universal pattern**

Pooling across models hides the effect



- **The average moves less than three of the five models**
- **Model × scale: three times the variance of scale alone**

The average describes no real model

Remark[3]

- **Rescaling the payoffs, an inert change of units in theory, moves cooperation in every model**

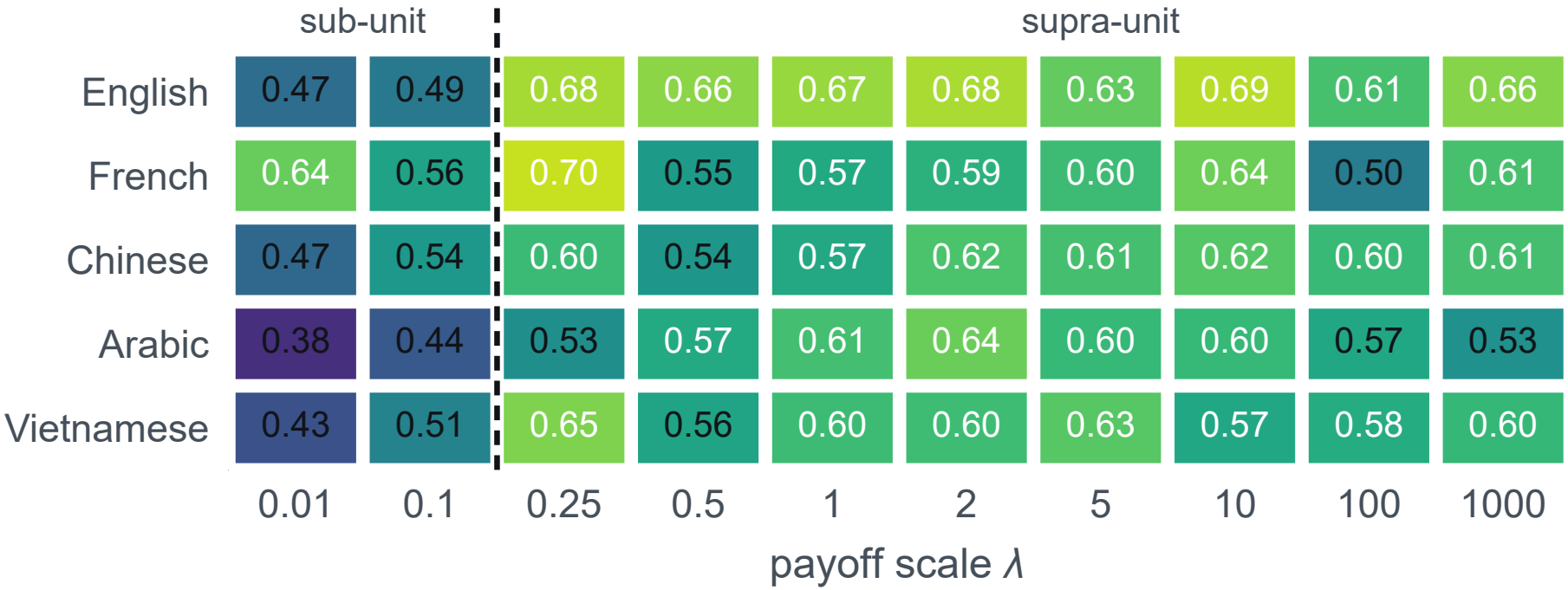
- **The response is model-specific in size, shape and direction**

- **Averages over models conceal it**

[3] Han, Huynh, Duy-Minh, …, Buscemi, Trang. **Payoff scaling shapes cooperation in LLM agents across languages**. arXiv:2601.19082, 2026.

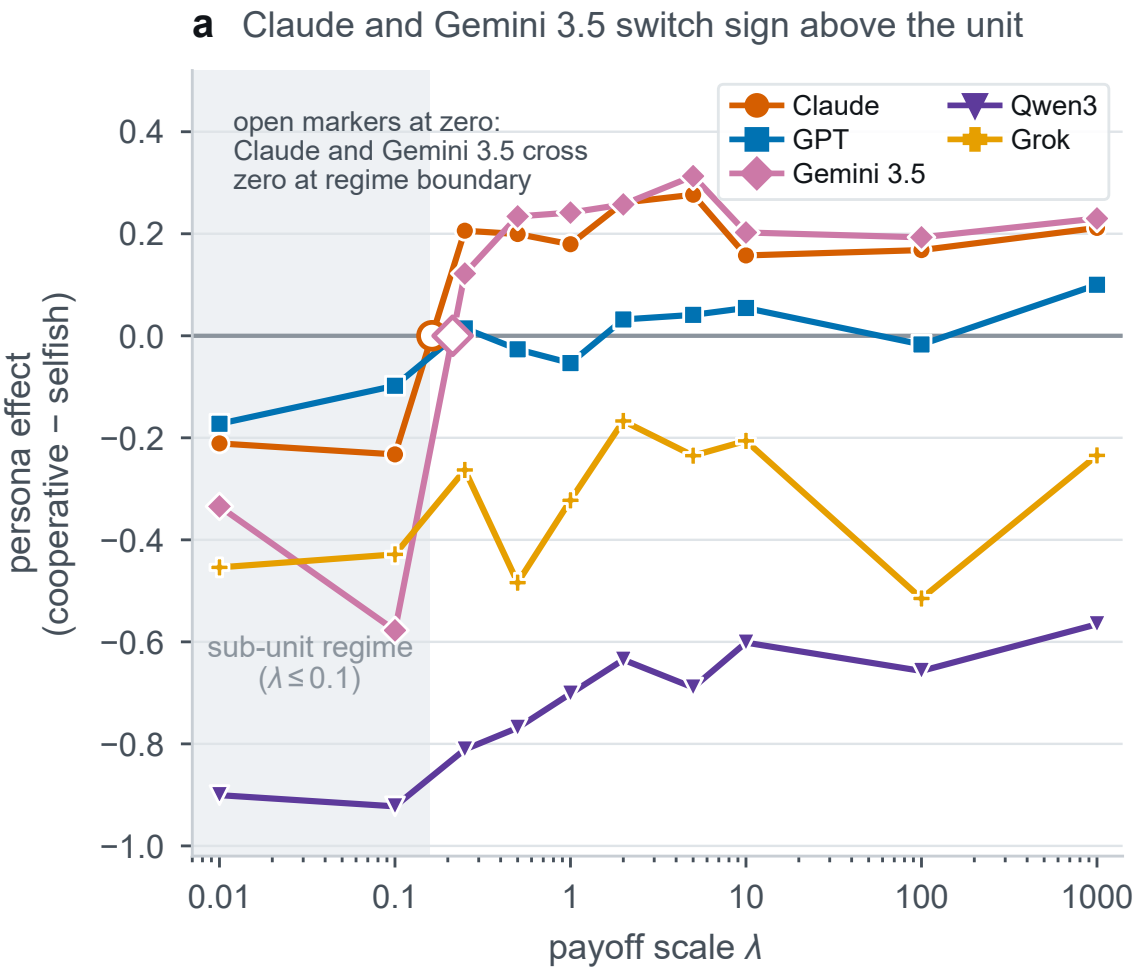
Scale sensitivity depends on language

a cooperation, language by scale



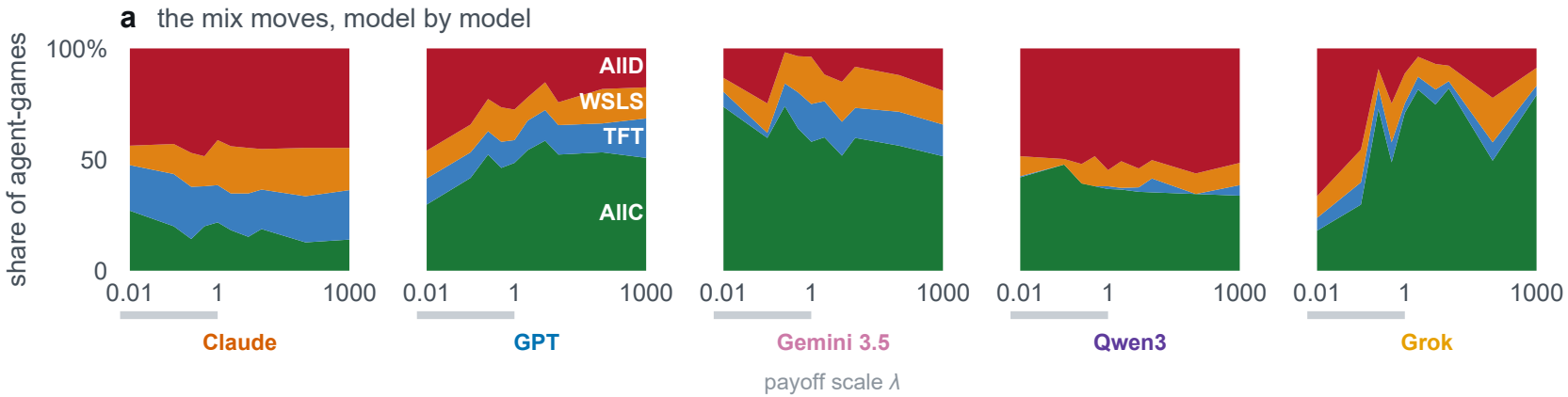
- **Within a model, sensitivity differs about two-fold between languages**
- **English is the most scale-sensitive language for 4 of 5 models**

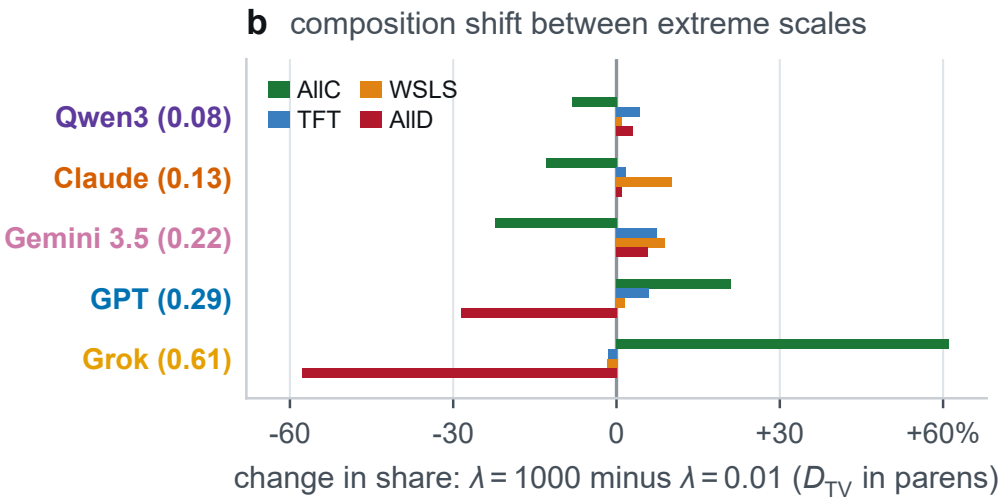
Payoff scale flips the persona instruction



- **At $\lambda \leq 0.1$, all five models cooperate more when told *"You are selfish"***
- **Claude and Gemini flip sign above it**
- **Qwen and Grok: inverted at every scale**

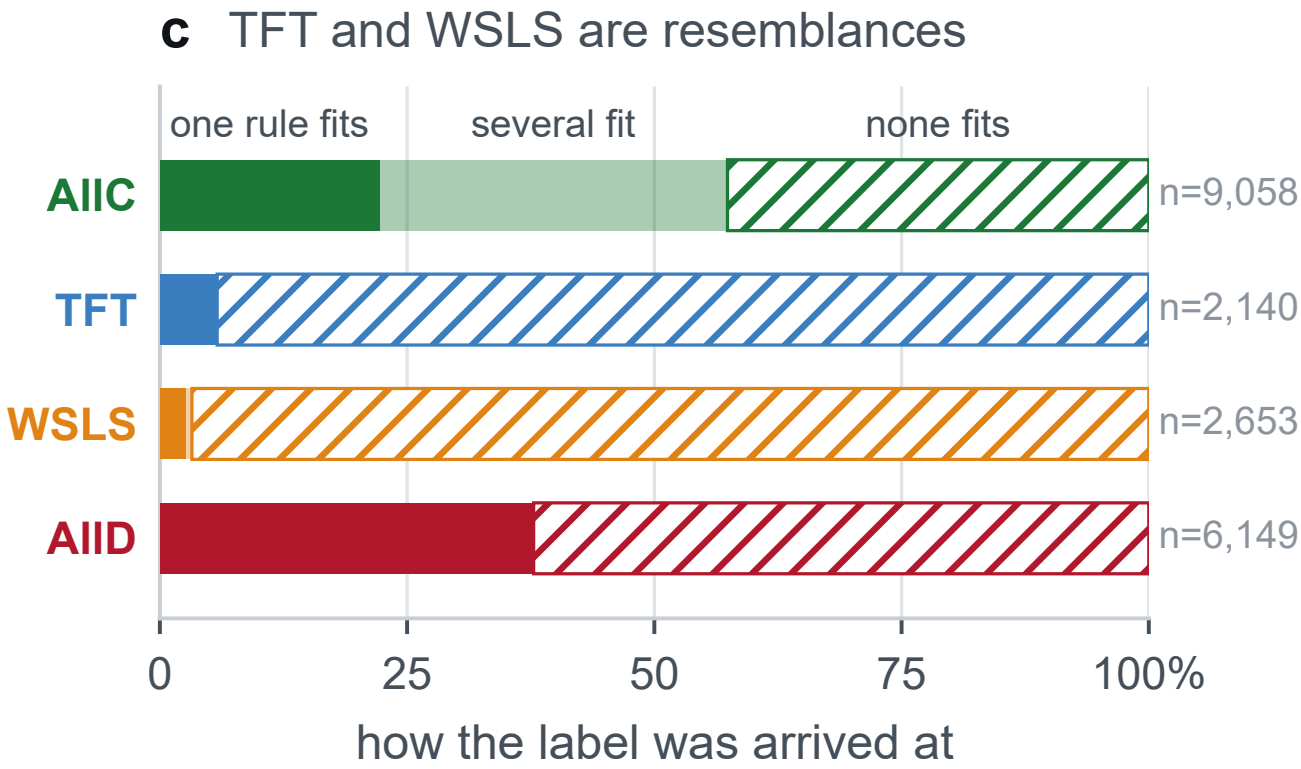
Payoff scale changes the strategy mix





- **GPT: AllD → AllC**
- **Grok: complete reversal**

Strategy labels are resemblances



- **95% of TFT and 97% of WSLS labels fit no rule exactly**
- **As stakes grow, play drifts away from the textbook rules in most models**

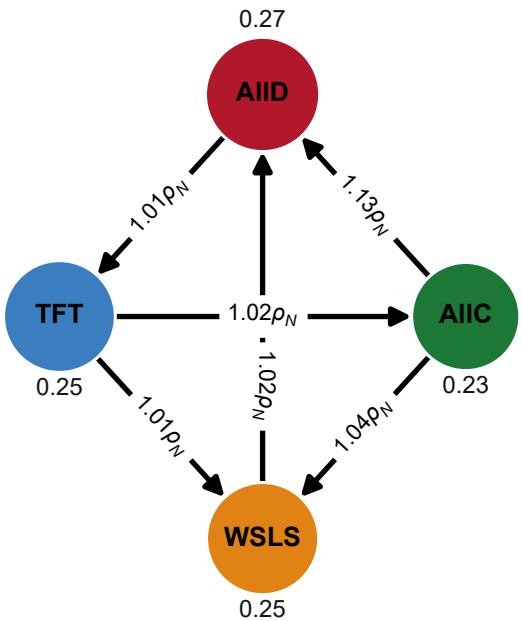
Where payoff scale does matter

- **Finite populations with social imitation:**

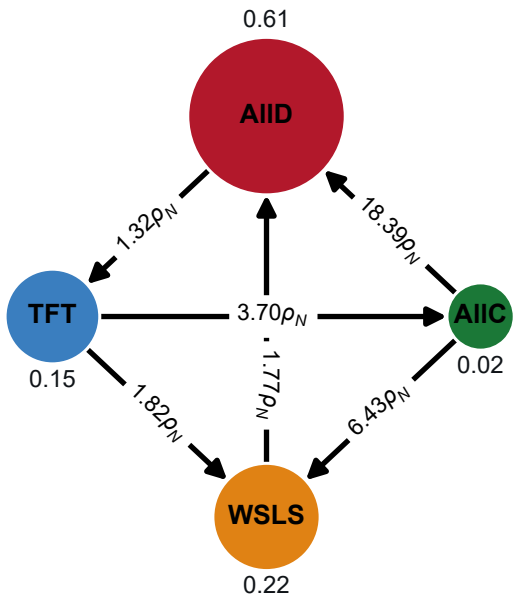
$$P(i \to j) = \frac{1}{1 + e^{-\beta\lambda\,(\pi_j - \pi_i)}}$$

- **Raising the stakes = raising the intensity of selection**

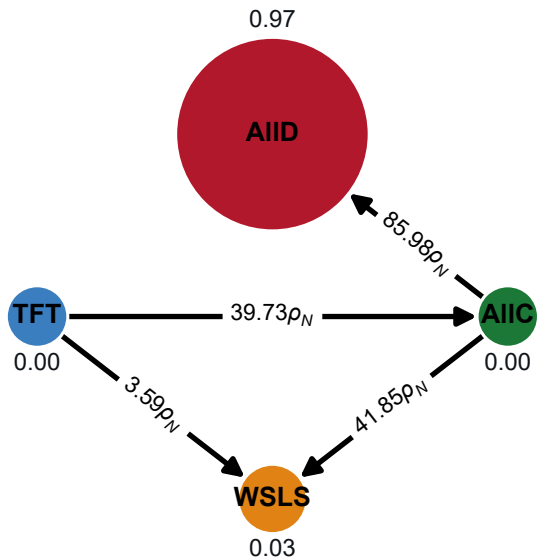
a $\lambda = 0.01$ nothing is selected



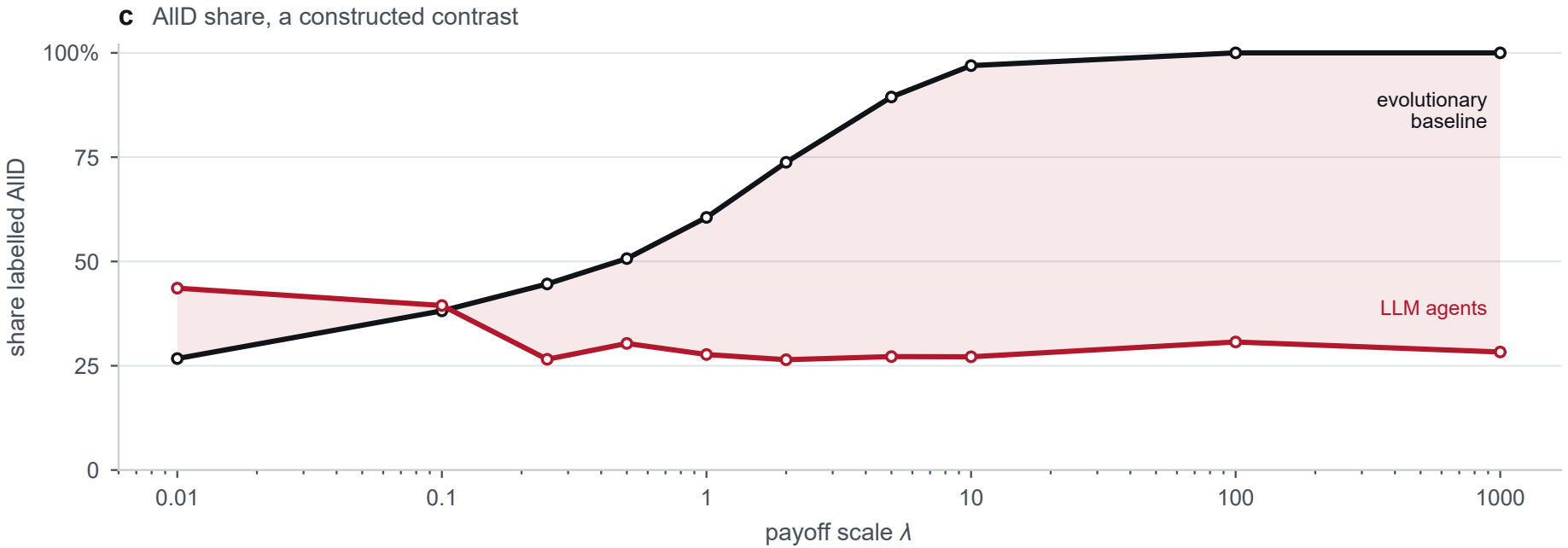
b $\lambda = 1$ the dilemma appears



c $\lambda = 10$ AllD takes 96.9%

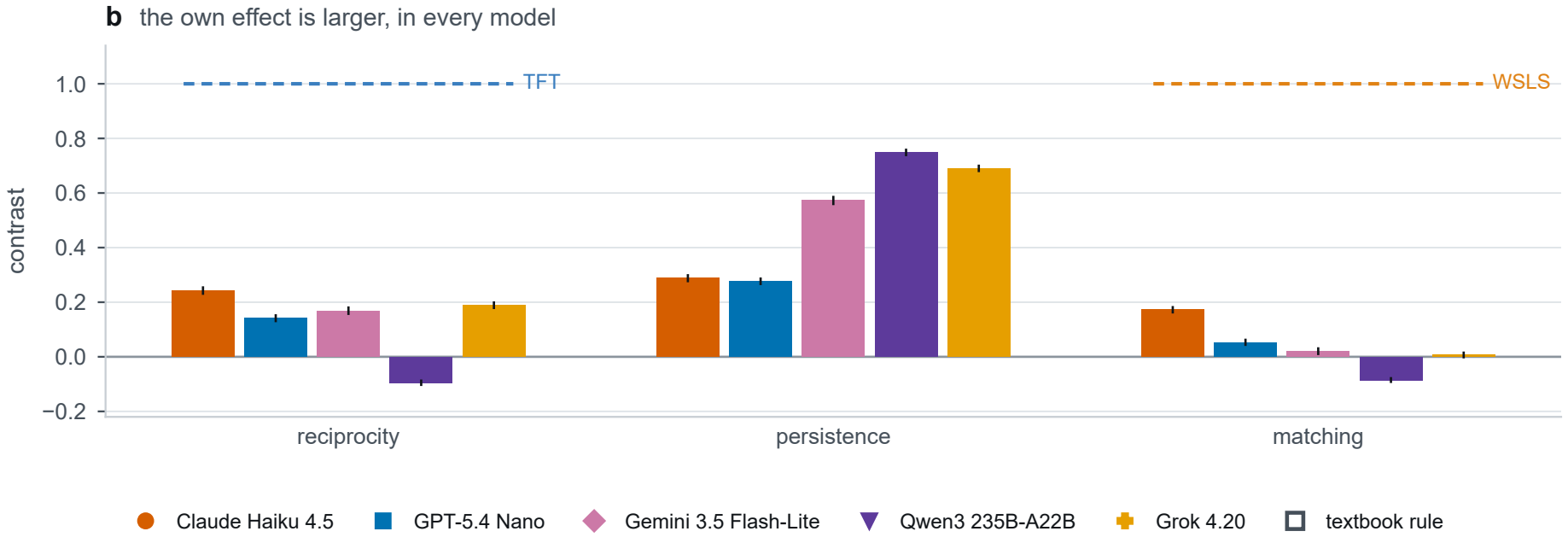


Pooled LLM agents move the opposite way



- **Evolutionary baseline: more AllD as stakes grow**
- **LLM agents, pooled over models: less AllD**

Agents condition on their own past moves



- **Persistence (own last move) beats reciprocity (partner's last move) in every model**

Remark

- **Payoff scale changes not only how often models cooperate, but also which rule their play resembles**

- **As stakes grow, pooled LLM agents move away from AllD, evolutionary dynamics towards it**

- **Agents respond more to their own moves than to their partner's**

Why? Four candidate explanations

1. **Prompt text that does not scale**
   narrative and persona
2. **How numbers are represented**
   sub-unit payoffs are rare and tokenised differently
3. **Provider-specific post-training**
4. **A distorted human stake response learned from text**

Not mutually exclusive; the design cannot separate them

Conclusions

- **Payoff scale, formally irrelevant, systematically changes machine cooperation**
- **The effect is model-specific, language-dependent, and changes what persona prompts do**
- **It also changes which strategy the play resembles**
- **A cooperation rate is not a model trait: report the payoff scale and test scale sensitivity**

Formal equivalences in the game are not automatically respected by the agents prompted to play it

Thank you!

QUESTIONS?

Payoff scaling shapes cooperation in LLM agents across languages

T. A. Han, T.-K. Huynh, D.-S. Duy-Minh, T.-B. Cao, P.-H. Le, H.-D. Nguyen, P.-Q. Nguyen-Lam, M.-L. Nguyen-Vo, H.-P. Pham, P.-H. Pham, T.-K. Than, C.-N. Tran, H. Tran, G.-T. Tran-Le, A. Buscemi, L. H. Trang

Paper

arXiv:2601.19082

Data and code

Zenodo

t.han@tees.ac.uk

Backup: how large is the departure?

	cooperation					Wald test	
Model	min	max	**range**	95% CI	p_{perm}	χ^2_9	p
Claude Haiku 4.5	0.371	0.452	**0.081**	[0.062, 0.135]	0.0025	32.3	$<$0.001
GPT-5.4 Nano	0.438	0.680	**0.243**	[0.200, 0.290]	0.0005	164.1	$<$0.001
Gemini 3.5 Flash-Lite	0.682	0.872	**0.190**	[0.165, 0.240]	0.0005	129.0	$<$0.001
Qwen3 235B-A22B	0.399	0.494	**0.095**	[0.068, 0.158]	0.0005	96.9	$<$0.001
Grok 4.20	0.274	0.910	**0.636**	[0.588, 0.686]	0.0005	1227.3	$<$0.001

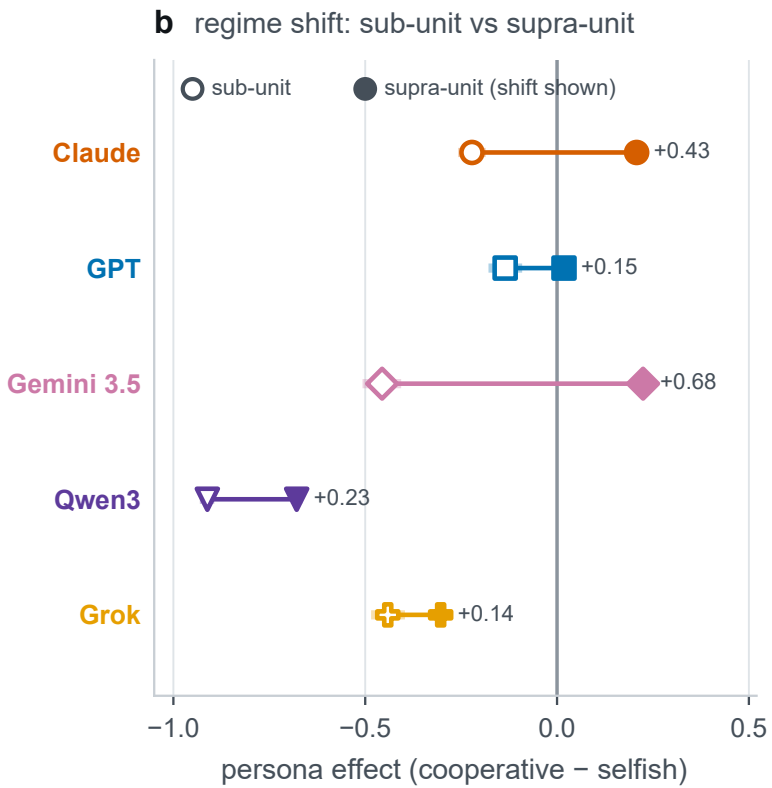
Range over the ten scales, CI from 1,500 whole-dyad bootstrap draws. Permutation test shuffles the scale label among whole dyads (2,000 draws; 0.0005 is the floor). Per-model logistic regression, scale as a 10-level factor, language and personas as covariates, SE clustered on the dyad.

Backup: variance decomposition

Term	Sum sq.	df	F	p	% variance
Model	4.7871	4	187.27	<0.001	45.4
Language	0.1599	4	6.25	<0.001	1.5
Payoff scale	0.6119	9	10.64	<0.001	5.8
Model $\times$ language	1.7901	16	17.51	<0.001	17.0
Model $\times$ scale	1.8910	36	8.22	<0.001	**17.9**
Language $\times$ scale	0.3844	36	1.67	0.018	3.6
Residual	0.9202	144			8.7

Type-II ANOVA on the 250 model $\times$ language $\times$ scale cell means; the residual is the omitted three-way interaction. Without Grok the scale main effect is not significant ($F_{9,108} = 1.15$, $p = 0.33$).

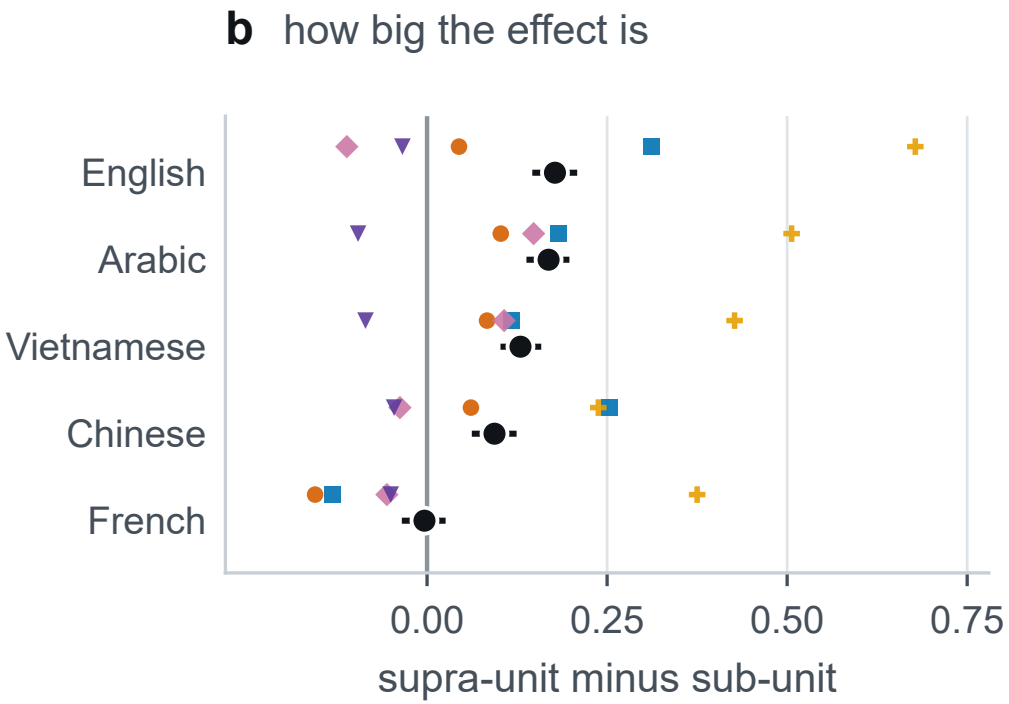
Backup: persona effect by regime



Model	$\lambda \leq 0.1$	$\lambda \geq 0.25$	flip
Claude	-0.222	$+0.207$	yes
GPT	-0.135	$+0.018^{\text{ns}}$	no
Gemini	-0.456	$+0.224$	yes
Qwen3	-0.911	-0.679	no
Grok	-0.441	-0.303	no

A flip needs both regime intervals (95% dyad bootstrap) to exclude zero on opposite sides. GPT's supra-unit interval straddles zero: abolished, not reversed.

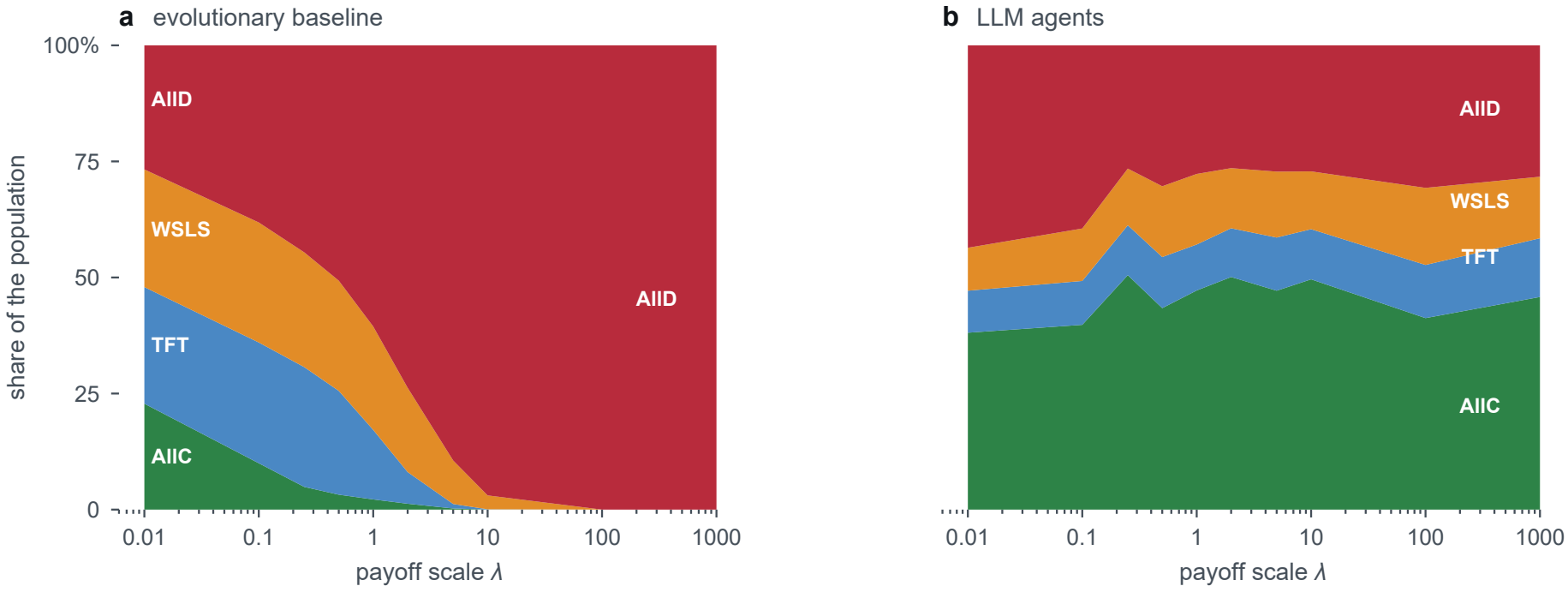
Backup: language effect by model



Supra-unit minus sub-unit cooperation. Black: five models pooled; markers: each model.

The Arabic and Chinese templates state penalties but ask to maximise a reward: a prompt effect, not a cultural one.

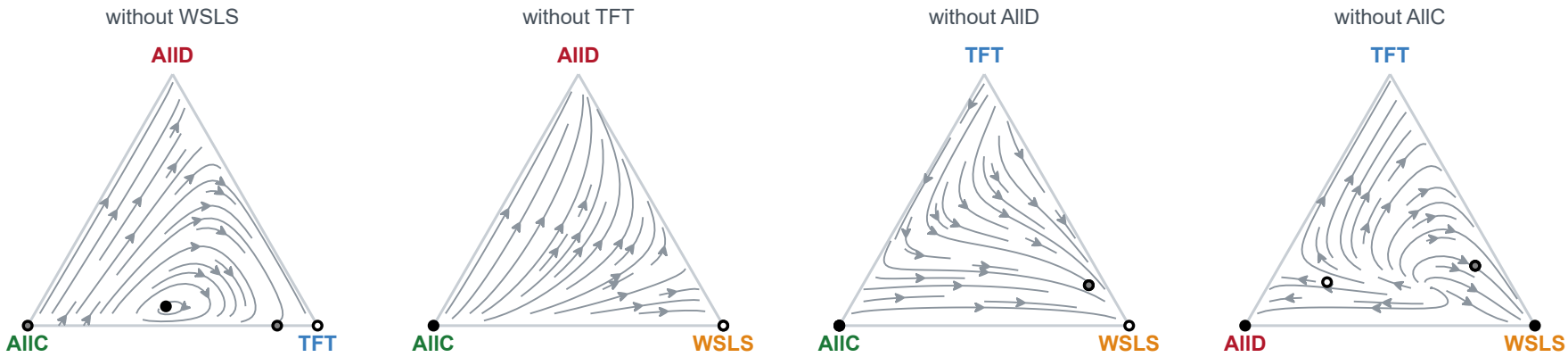
Backup: baseline and corpus, full mix



Left: stationary distribution of the evolutionary model ($Z = 100$, $\beta = 0.1$, $\varepsilon = 0.05$). Right: label shares in the corpus, pooled over models, languages and personas.

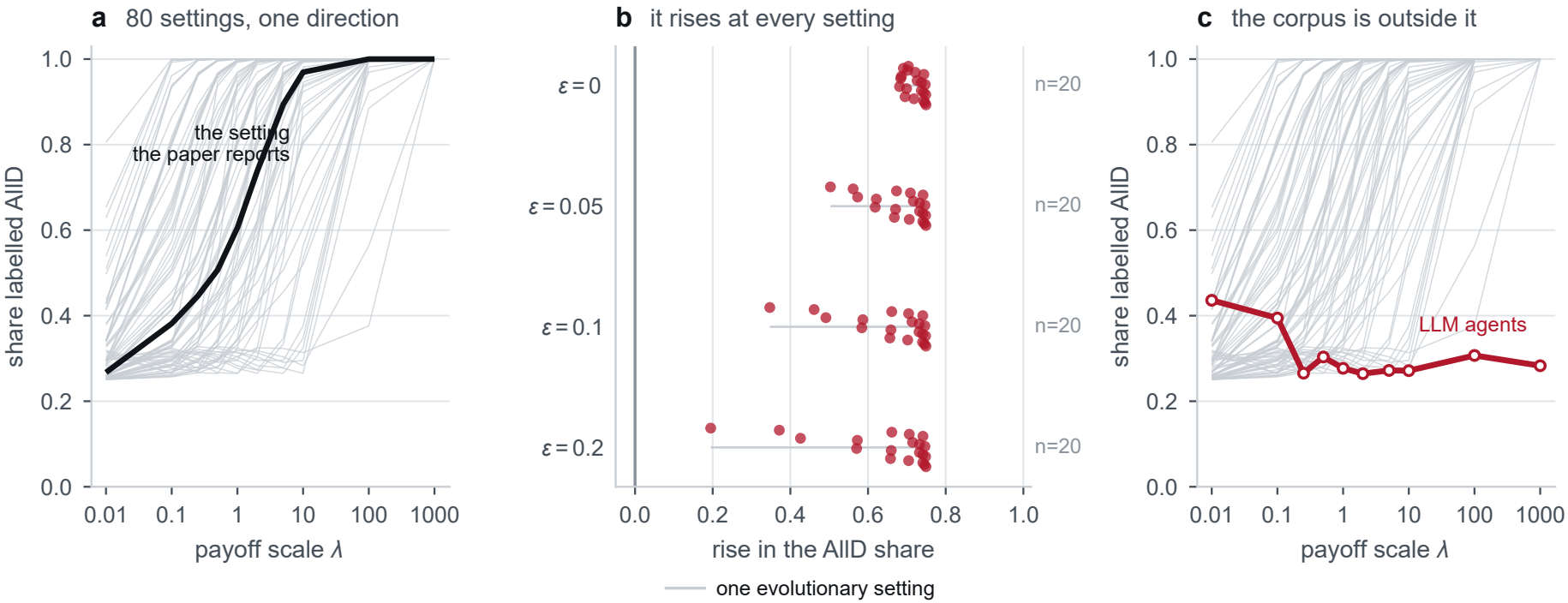
Backup: replicator dynamics ignore the scale

a the deterministic dynamics do not see the payoff scale at all



Normalised vector fields at $\lambda = 0.01$ and $\lambda = 1000$ agree to within 4×10^{-14} over 2,000 random population states: the scale changes the speed, never the path.

Backup: robustness of the evolutionary direction



80 settings, $Z \in \{50, 100, 200, 500\} \times \beta \in \{0.01, 0.05, 0.1, 0.5, 1\} \times \varepsilon \in \{0, 0.05, 0.1, 0.2\}$**: the stationary AllD share is higher at** $\lambda = 1000$ **than at** $\lambda = 0.01$ **in every one.**

Backup: limitations

- **Five models, one repeated game, one family of prompt templates**
- **Language partly confounded with template wording**
- **Ten rounds limit strategy inference**
- **No numeric-free condition: next, a verbal baseline describing each outcome only by its rank**

Image credits

Photographs from Wikimedia Commons.

- Robonaut and astronaut handshake: R. Markowitz, NASA, public domain
- Police interrogation room (movie set): nesnad, CC BY 3.0
- John von Neumann: courtesy of Los Alamos National Laboratory (free use with attribution)
- Prisoner's Dilemma sketch: Giulia Forsythe, CC0
- Tragedy and Comedy masks: Tim Green, CC BY 2.0
- Robot illustrations and model logos: the paper's Figure 1
- University logos: the universities' websites and Wikimedia Commons